

Week 8 — Mechanical Systems Diagnosis

Sessions: 1 (Theory) · 2 (Lab) · **Total:** 4 hours

Primary LOs: LO-10

Hands-on target: Session 2 ≈ 85% practical

Instructor Preparation Guide (Complete Day Before)

Pre-session setup

- [] Select carts with diverse mechanical wear if possible (brakes, play in steering, tire wear)
- [] Jack stands rated for cart; wheel chocks; never rely on jack alone
- [] Brake measurement tools; brake clean (low-VOC as policy); catch pans
- [] Gear oil for transaxle check; correct fluid type card
- [] Print labs/W08_mechanical_inspection.md and road-test sheet section
- [] Mark safe road-test loop; radios/spotter rules
- [] Note regen vs mechanical brake interaction for electric carts

Conditions to highlight (not all must be planted)

Condition	Teaching value
Low tire pressure one side	Pull / range loss
Parking brake out of adjustment	Safety fail
Wet/contaminated brakes	Noise / poor braking
Torn CV/axle boot or seepage (if equipped/visible)	Leak awareness
Excess steering play	Kingpin/tie rod inspection

Safety

- Hydraulic brake fluid: paint/eye hazard; use correct fluid only.
 - Lift points per OEM; jack stands before going under.
 - Road tests: helmets if shop policy; max training speed; instructor approval.
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Week 8 Learning Objectives

1. Inspect and diagnose brake, suspension, steering, and drivetrain issues.
 2. Explain interaction between regenerative braking and mechanical brakes.
 3. Perform road tests to identify handling, noise, and vibration problems.
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Session 1 — Theory & Demonstration (2 hours)

Materials needed: Brake component samples/photos; tire wear photo set; gear oil sample jars (labeled); jack/stand demo; regen-vs-friction whiteboard; road-test note example.

Timed Agenda

Time	Min	Activity	Instructor Notes
0:00	5	Safety: lift, crush, brake dust/fluid	
0:05	20	Brake systems: mechanical cable, hydraulic, regen blending	[DIAGRAM PLACEHOLDER: regen + mechanical brake blend concept]
0:25	15	Common brake complaints mapping	Pull, noise, low pedal, parking brake fail
0:40	15	Suspension: springs, shocks, bushings	Wear indicators
0:55	5	Break	
1:00	15	Steering: rack & pinion / tie rods / kingpins; play checks	
1:15	15	Transaxle, bearings, seals, gear oil	Leak vs seepage triage
1:30	15	Tires/wheels: pressure critical for range & handling	Wear patterns
1:45	10	Road-test methodology & note-taking	
1:55	5	Lab preview	

Key Teaching Points

- Regen reduces mechanical brake wear but does **not** replace a working friction brake system.
- Tire pressure is a “range complaint” root cause often missed.
- Noise diagnosis: reproduce safely, localize front/rear/left/right, load vs coast.
- Parking brake must hold on shop’s designated grade test.

Common Misconceptions

Misconception	Correction
“Electric carts don’t need brake maintenance.”	Mechanical brakes still critical for stops/parking.
“All pulls are alignment.”	Brakes, tires, and bent components also pull.
“Topping off gear oil fixes bearing noise.”	Find leak source; noise may mean imminent failure.

Safety Reminders

- Never work under cart supported only by jack.

- Control dust; avoid compressed-air blasting of brake dust toward face.
- After brake work: pump/adjust per system; test at low speed first.

Session 2 — Hands-On Lab (2 hours)

Materials needed: labs/W08_mechanical_inspection.md; jack stands; brake measuring tools; tire gauge; flashlights; road-test loop marked; spotter vests if required by shop policy.

Timed Agenda

Time	Min	Activity	Instructor Notes
0:00	5	PPE; lift briefing; assign carts	
0:05	25	Brake inspection: lining thickness, hardware, cables/hydraulics, parking brake	Measure & record
0:30	20	Steering & suspension play checks (safely supported)	Instructor verifies stand placement
0:50	5	Break	
0:55	15	Transaxle oil level/condition & leak inspection	
1:10	25	Road test (pairs: driver + observer notes)	Use checklist; low speed
1:35	15	Complete mechanical report; discuss findings	
1:50	10	Lower carts; torque wheel lugs; cleanup	

Assessment Focus

Mechanical inspection checklist + road test observation notes.

Week 8 Quiz (8 questions)

- Regenerative braking:
 - Eliminates need for parking brakes
 - Converts some kinetic energy electrically and works with mechanical brakes
 - Waters batteries
 - Aligns steering
- A cart that pulls left under braking may have:
 - Only a software update need
 - Contaminated/misadjusted brakes, tire issues, or mechanical defects — needs systematic inspection
 - Too much windshield fluid
 - A good alignment by definition

3. Before going under a raised cart:
 - A) Trust the jack alone
 - B) Use rated jack stands on proper lift points + chocks
 - C) Have a coworker sit in the seat
 - D) Remove all stands for access
4. Tire pressure affects:
 - A) Only cosmetics
 - B) Handling, tire life, and effective range
 - C) Controller flash color
 - D) Spark plug gap
5. True or False: Parking brake performance should be verified as part of mechanical inspection.

Short Answer

1. List three items checked during a steering play inspection.
2. What gear oil observations would concern you?
3. Why use a driver + observer during training road tests?

Answer Key

1. **B**
2. **B**
3. **B**
4. **B**
5. **True**
6. Tie rod ends, kingpins/bushings, rack mounts, wheel bearings, etc.
7. Low level, burnt smell, metal glitter, milky (water), active leaks at seals
8. One focuses on vehicle control/safety; one records noises, pulls, conditions, speeds